

# SELECT from WORLD Tutorial

|           |                    |
|-----------|--------------------|
| Language: | English • 日本語 • 中文 |
|-----------|--------------------|

| name        | continent | area    | population | gdp          |
|-------------|-----------|---------|------------|--------------|
| Afghanistan | Asia      | 652230  | 25500100   | 20343000000  |
| Albania     | Europe    | 28748   | 2831741    | 12960000000  |
| Algeria     | Africa    | 2381741 | 37100000   | 188681000000 |
| Andorra     | Europe    | 468     | 78115      | 3712000000   |
| Angola      | Africa    | 1246700 | 20609294   | 100990000000 |
| ...         |           |         |            |              |

In this tutorial you will use the SELECT command on the table world:

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## Introduction

# 1.

Read the notes about this table. Observe the result of running this SQL command to show the name, continent and population of all countries.

```
SELECT name, continent, population FROM world;
```

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|                               |               |            |
|-------------------------------|---------------|------------|
| Burundi                       | Africa        | 9420248    |
| Cambodia                      | Asia          | 15184116   |
| Cameroon                      | Africa        | 20386799   |
| Canada                        | North America | 35427524   |
| Cape Verde                    | Africa        | 491875     |
| CÃ'te d'Ivoire                | Africa        | 23919000   |
| Central African Republic      | Africa        | 4709000    |
| Chad                          | Africa        | 13211000   |
| Chile                         | South America | 17773000   |
| China                         | Asia          | 1365370000 |
| Colombia                      | South America | 47662000   |
| Comoros                       | Africa        | 743798     |
| Congo, Democratic Republic of | Africa        | 69360000   |
| Congo, Republic of            | Africa        | 4559000    |

## Large Countries

# 2.

How to use WHERE to filter records. Show the name for the countries that have a population of at least 200 million. 200 million is 200000000, there are eight zeros.

```
SELECT name FROM world
WHERE population >=200000000;
```

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## Correct answer

| name          |
|---------------|
| Brazil        |
| China         |
| India         |
| Indonesia     |
| United States |

## Per capita GDP

# 3.

Give the name and the **per capita GDP** for those countries with a population of at least 200 million.

*HELP:How to calculate per capita GDP*

```
SELECT name, gdp / population AS per_capita_gdp
FROM world
WHERE population >= 200000000;
```

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## Correct answer

| name          | per_capita_gdp |
|---------------|----------------|
| Brazil        | 11115.2648     |
| China         | 6121.7106      |
| India         | 1504.7931      |
| Indonesia     | 3482.0205      |
| United States | 51032.2945     |

sfwQ3



## South America In millions

4.

Show the name and population in millions for the countries of the continent 'South America'. Divide the population by 1000000 to get population in millions.

```
SELECT name, population / 1000000 AS population_in_millions
FROM world
WHERE continent = 'South America';
```

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## Correct answer

| name      | population_in_millions |
|-----------|------------------------|
| Argentina | 42.6695                |
| Bolivia   | 10.0273                |
| Brazil    | 202.7940               |
| Chile     | 17.7730                |
| Colombia  | 47.6620                |
| Ecuador   | 15.7742                |
| Guyana    | 0.7849                 |
| Paraguay  | 6.7884                 |

## France, Germany, Italy

5.

Show the name and population for France, Germany, Italy

```
SELECT name, population
FROM world
WHERE name IN ('France', 'Germany', 'Italy');
```

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## Correct answer

| name    | population |
|---------|------------|
| France  | 65906000   |
| Germany | 80716000   |
| Italy   | 60782668   |

## United

# 6.

Show the countries which have a name that includes the word 'United'

```
SELECT name
FROM world
WHERE name LIKE '%United%';
```

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## Correct answer

| name                 |
|----------------------|
| United Arab Emirates |
| United Kingdom       |
| United States        |

## Two ways to be big

7.

Two ways to be big: A country is **big** if it has an area of more than 3 million sq km or it has a population of more than 250 million.

**Show the countries that are big by area or big by population. Show name, population and area.**



```
SELECT name, population, area
FROM world
WHERE area > 3000000
OR population > 250000000;
```

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## Correct answer

| name           | population | area     |
|----------------|------------|----------|
| Australia      | 23545500   | 7692024  |
| Brazil         | 202794000  | 8515767  |
| Canada         | 35427524   | 9984670  |
| China          | 1365370000 | 9596961  |
| India          | 1246160000 | 3166414  |
| Indonesia      | 252164800  | 1904569  |
| Russia         | 146000000  | 17125242 |
| United Kingdom | 61000000   | 242467   |

## One or the other (but not both)

8.

**Exclusive OR (XOR). Show the countries that are big by area (more than 3 million) or big by population (more than 250 million) but not both. Show name, population and area.**

- Australia has a big area but a small population, it should be **included**.
- Indonesia has a big population but a small area, it should be **included**.
- China has a big population **and** big area, it should be **excluded**.
- United Kingdom has a small population and a small area, it should be **excluded**.

```
SELECT name, population, area
FROM world
WHERE (area > 3000000 AND population <= 250000000)
      OR (area <= 3000000 AND population > 250000000);
```

[Submit SQL](#)[restore default](#)

## Correct answer

| name      | population | area     |
|-----------|------------|----------|
| Australia | 23545500   | 7692024  |
| Brazil    | 202794000  | 8515767  |
| Canada    | 35427524   | 9984670  |
| Indonesia | 252164800  | 1904569  |
| Russia    | 146000000  | 17125242 |

## Rounding

# 9.

Show the name and population in millions and the GDP in billions for the countries of the continent 'South America'. Use the ROUND function to show the values to two decimal places.

**For Americas show population in millions and GDP in billions both to 2 decimal places.**

*Millions and billions*

*Missing decimals*

```
SELECT
  name,
  ROUND(population / 1000000, 2) AS population_in_millions,
  ROUND(gdp / 10000000000, 2) AS gdp_in_billions
FROM world
WHERE continent = 'South America';
```

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## Correct answer

| name      | population_in_millions | gdp_in_billions |
|-----------|------------------------|-----------------|
| Argentina | 42.67                  | 477.03          |
| Bolivia   | 10.03                  | 27.04           |
| Brazil    | 202.79                 | 2254.11         |
| Chile     | 17.77                  | 268.31          |
| Colombia  | 47.66                  | 369.81          |
| Ecuador   | 15.77                  | 87.50           |
| Guyana    | 0.78                   | 2.85            |

Play the SELECT card game (<https://sqlzoo.net/40289347/>)

1:14



SCORE: 5

```
SELECT card
FROM cards
WHERE number = 1
AND colour = 'red'
```

SELECT Card Game

## Trillion dollar economies

# 10.

Show the name and per-capita GDP for those countries with a GDP of at least one trillion (1000000000000; that is 12 zeros). Round this value to the nearest 1000.

Show per-capita GDP for the trillion dollar countries to the nearest \$1000.

```
SELECT
  name,
  ROUND(gdp / population, -3) AS per_capita_gdp
FROM world
WHERE gdp >= 1000000000000;
```

Submit SQL[restore default](#)

## Correct answer

| name      | per_capita_gdp |
|-----------|----------------|
| Australia | 66000          |
| Brazil    | 11000          |
| Canada    | 45000          |
| China     | 6000           |
| France    | 40000          |
| Germany   | 42000          |
| India     | 2000           |
| U.S.      | 33000          |

## Name and capital have the same length

# 11.

Greece has capital Athens.

Each of the strings 'Greece', and 'Athens' has 6 characters.

**Show the name and capital where the name and the capital have the same number of characters.**

- You can use the LENGTH function to find the number of characters in a string

For Microsoft SQL Server the function LENGTH is LEN

```
SELECT name, capital
FROM world
WHERE LENGTH(name) = LENGTH(capital);
```

Submit SQL

restore default

## Correct answer

| name     | capital  |
|----------|----------|
| Algeria  | Algiers  |
| Angola   | Luanda   |
| Armenia  | Yerevan  |
| Botswana | Gaborone |
| Canada   | Ottawa   |
| Djibouti | Djibouti |
| Egypt    | Cairo    |
| France   | Paris    |

## Matching name and capital

# 12.

The capital of Sweden is Stockholm. Both words start with the letter 'S'.

**Show the name and the capital where the first letters of each match. Don't include countries where the name and the capital are the same word.**

- You can use the function LEFT to isolate the first character.
- You can use <> as the NOT EQUALS operator.

```
SELECT name, capital
FROM world
WHERE LEFT(name, 1) = LEFT(capital, 1)
AND name <> capital;
```

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## Correct answer

| name     | capital             |
|----------|---------------------|
| Algeria  | Algiers             |
| Andorra  | Andorra la Vella    |
| Barbados | Bridgetown          |
| Belize   | Belmopan            |
| Brazil   | Brasília            |
| Brunei   | Bandar Seri Begawan |
| Burundi  | Bujumbura           |
| Cameroon | Yaoundé             |

## All the vowels

# 13.

**Equatorial Guinea** and **Dominican Republic** have all of the vowels (a e i o u) in the name. They don't count because they have more than one word in the name.

**Find the country that has all the vowels and no spaces in its name.**

- You can use the phrase `name NOT LIKE '%a%'` to exclude characters from your results.
- The query shown misses countries like Bahamas and Belarus because they contain at least one 'a'

```
SELECT name
FROM world
WHERE name LIKE '%a%'
      AND name LIKE '%e%'
      AND name LIKE '%i%'
      AND name LIKE '%o%'
      AND name LIKE '%u%'
      AND name NOT LIKE '% %';
```

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## Correct answer

| name       |
|------------|
| Mozambique |

## What Next

- BBC QUIZ
- You can to continue practising the the same techniques and gain more experience of the basic skills on the Nobel table. The WHERE statement using the nobel table.
- You can learn about nested statements, these are instructive and entertaining, but not essential for beginners. Nested SELECT statements using the world table.

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